Assignment 2 - AE4238

Aero Engine Technology

Combustor Analysis & Performance

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1 Problem Description

The presented combustion chamber belongs to a Rolls-Royce TRENT XWB-84 aero engine. As any other combustion chamber, its function is to raise the temperature of the incoming high-pressure air flow by burning fuel in the most efficient manner possible. The design of such a device should aim toward minimizing NO_x and CO emissions while keeping to its target temperatures.

In the pages that follow a report is developed, performing a comprehensive analysis of the combustion process and the device itself. It covers performance at various thrust conditions, fuel properties and air flow distributions, among others.

2 Assumptions

In order to make the analysis more straight forward, the following assumptions are applied, enumerated alongside the ones provided:

1. **Steady State:** The engine is assumed to operate under steady-state conditions for all points.

2. **Adiabatic Device:** The combustor is adiabatic; heat loss to the surroundings is considered negligible and will thus not be taken into account.

3. **Ideal Gas:** The working fluid behaves as an ideal gas following $P = \rho RT$.

4. **Combustion Efficiency:** For all calculations, the efficiency is assumed $\eta_{cc} = 100\%$, to simplify calculations, as it will not harm accuracy noticeably.

5. **Specific heats:** $C_{p,air} = 1000\,\text{J/kgK}$ and $C_{p,gas} = 1150\,\text{J/kgK}$.

6. **Partial Oxidation:** In fuel-rich zones ($\phi > 1$), it is assumed that the products of combustion are water vapour (H_2O) and Carbon Monoxide (CO) instead of H_2O and CO_2.

7. **Combustor Geometry:** The combustor geometry is annular, and the entire combustor is assumed to be split into 18 sections.

8. **Zonal Definitions:** The ϕ for a given zone is based on the fuel supplied to the combustor and the amount of air present in that zone. Thus, the Fuel-to-Air Ratio (FAR) is defined as $\frac{\dot{m}_f}{0.12 \cdot \dot{m}_3 + X_1}$ for the Rich Zone and $\frac{\dot{m}_f}{X_2}$ for the Quench Zone.

3 Procedure

3.1 Point 1: Fuel Calorific Value

The fuel employed in the engine is a Sustainable Aviation Fuel (SAF) with formula $C_{11}H_{23}$. To obtain the Lower Heating Value (LHV), we must first determine the stoichiometric coefficients (here named x, y, z) for the oxidation reaction (air taken with proportions $79\%N_2$, $21\%O_2$):

$$C_{11}H_{23} + x(O_2 + 3.76N_2) \rightarrow yCO_2 + zH_2O + x3.76N_2 \tag{1}$$

First we solve for the unknowns by balancing the species on both sides, by means of simple algebra:

- **Carbon (C):** $11 = y \implies y = 11$
- **Hydrogen (H):** $23 = 2z \implies z = 11.5$
- **Oxygen (O):** $2x = 2y + z$

Which substituting y and z into the oxygen equation yields:

$$2x = 2(11) + 11.5 = 22 + 11.5 = 33.5 \implies x = 16.75 \tag{2}$$

The balanced equation is:

$$C_{11}H_{23} + 16.75(O_2 + 3.76N_2) \rightarrow 11CO_2 + 11.5H_2O + 62.98N_2 \tag{3}$$

The next step is to apply Hess's Law - The heat of combustion (ΔH_c) is calculated using standard enthalpies of formationΔH_f° listen in the assignment statement:

$$\Delta H_c = \sum (n\Delta H_f^\circ)_{\text{prod}} - \sum (n\Delta H_f^\circ)_{\text{react}} \tag{4}$$

Finally, the LHV is the energy per unit mass, and so we simply divide by the molar mass of the fuel:

$$LHV = \frac{|\Delta H_c|}{M_{fuel}} \tag{5}$$

3.2 Point 2: Fuel Flow Rate

Considering the combustion chamber inner surface as control volume, we can apply the energy conservation principle:

$$\dot{m}_{air}h_{air}(T_3) + \dot{m}_f \cdot LHV \cdot \eta_{cc} = (\dot{m}_{air} + \dot{m}_f)h_{gas}(T_4) \tag{6}$$

Considering constant specific heats (taking enthalpy as $h = C_pT$) and solving for $\dot{m}_f$:

$$\dot{m}_f = \dot{m}_{air}\frac{C_{p,gas}T_4 - C_{p,air}T_3}{LHV - C_{p,gas}T_4} \tag{7}$$

3.3 Point 3: Overall Equivalence Ratio

The overall equivalence ratio is defined as to express the global leanness or richness of the fuel and air mixture of the chamber. It is written as:

$$\phi_{overall} = \frac{FAR_{actual}}{FAR_{stoich}} \tag{8}$$

To derive the stoichiometric fuel-to-air Ratio (FAR_{stoich}) we must first acknowledge it being a constant property of the fuel, derived from the balanced chemical equation in the first point:

$$C_{11}H_{23} + 16.75(O_2 + 3.76N_2) \rightarrow 11CO_2 + 11.5H_2O + 16.75(3.76)N_2 \tag{9}$$

Which means that for one mole of fuel ($M_{fuel} = 155.184\,\text{g/mol}$), we need 16.75 moles of oxygen ($M_{O_2} = 32.00\,\text{g/mol}$). The total mass of oxygen is:

$$Mass_{O_2,stoich} = 16.75 \times 32.00 = 536.00\,\text{g} \tag{10}$$

and if we assume that standard air contains a percentage of 23.2% oxygen by mass ($Y_{O_2} \approx 0.232$), then the total mass of air needed will be:

$$Mass_{Air,stoich} = \frac{Mass_{O_2,stoich}}{Y_{O_2}} = \frac{536.00}{0.232} \approx 2310.34\,\text{g} \tag{11}$$

Finally, the stoichiometric fuel-to-air ratio is:

$$FAR_{stoich} = \frac{M_{fuel}}{Mass_{Air,stoich}} = \frac{155.184}{2310.34} \approx 0.0672 \tag{12}$$

and now the equivalence ratio can be computed directly from the provided data.

3.4 Point 4: Air Flow Distribution

To calculate the air mass flow required for each zone (Rich - Primary zone, Quench - Secondary zone, Lean - Dilution zone), we derive the formula relating mass flow to the equivalence ratio by means of the fuel-to-air ratio.

Rich Zone ($\phi = 1.5$): As we assumed, combustion in the rich zone will produce CO instead of CO_2. A partial oxidation will require less oxygen (when re-doing the balance it is 11.25 moles instead of 16.75), leading to a higher stoichiometric FAR:

$$FAR_{stoich,partial} = FAR_{stoich} \times \frac{16.75}{11.25} \approx 0.1001 \tag{13}$$

Using this adjusted value, he rich zone air mass flow is calculated:

$$\dot{m}_{air,rich} = \frac{\dot{m}_f}{\phi_{rich} \times FAR_{stoich,partial}} \tag{14}$$

Quench Zone ($\phi = 0.7$): Since the rich zone is by definition fuel rich ($\phi > 1$), only a portion of the fuel will react, as it is limited by the available air. The remaining unburnt fuel must be processed in the quench zone:

$$\dot{m}_{fuel,reacted} = \dot{m}_{air,rich} \times FAR_{stoich,partial} \tag{15}$$

$$\dot{m}_{fuel,unburnt} = \dot{m}_{f,total} - \dot{m}_{fuel,reacted} \tag{16}$$

The air added by the quench holes is calculated to dilute specifically this *unburnt* fuel to the target lean condition ($\phi = 0.7$), where complete combustion (standard FAR_{stoich}) occurs:

$$\dot{m}_{air,added\_Q} = \frac{\dot{m}_{fuel,unburnt}}{\phi_{quench} \times FAR_{stoich}} \tag{17}$$

The remaining mass rate is found in the dilution zone, where no combustion takes place, so it will just be the remaining flow rate off the total.

3.5 Point 5: Adiabatic Flame Temperatures

The adiabatic flame temperature is best obtained by accounting for the temperature dependence of specific heat capacity $C_p(T)$. For this we have opted for the approach of using **NIST Shomate polynomials**, from a WebBook. Since the enthalpy functions are non-linear, T_{ad} is determined by iteratively solving the following energy balance through a MATLAB code:

$$\sum \dot{m}_{in} h(T_{in}) + \dot{m}_f \cdot LHV_{eff} = \sum \dot{m}_{out} h(T_{ad}) \tag{18}$$

Where sensible enthalpy $h(T)$ is calculated via polynomial integration of C_p relative to 298.15 K (the numerical root-finding algorithm MATLAB `fzero`):

$$h(t) = \left[A \cdot t + B\frac{t^2}{2} + C\frac{t^3}{3} + D\frac{t^4}{4} - \frac{E}{t} \right] \cdot \frac{1000}{M} \quad [\text{J/kg}] \tag{19}$$

where $t = T/1000$ and M is the molar mass of the species.

In the **Rich Zone**, partial oxidation is assumed. The reaction forms CO instead of CO_2, significantly reducing the heat release ($LHV_{rich} \approx 23.58\,\text{MJ/kg}$, obtained through the same procedure as before, this time balancing with CO).

In the **Quench and Lean Zones**, with the introduction of secondary air, complete combustion is assumed to take place. All carbon fully oxidizes to CO_2 (Standard $LHV \approx 43.60\,\text{MJ/kg}$).

3.6 Point 6: Combustor Volume

The combustor should be sized based on the maximum mass flow condition (Take-off/300 kN) to ensure a residence time (here denoted τ) of 6 ms. The volume can be related to the volumetric flow rate as follows:

$$V_{comb} = \dot{V}_{gas} \cdot \tau = \frac{\dot{m}_{total}}{\rho_{avg}} \cdot \tau \tag{20}$$

Where we declare an average density of the gas passing through the combustor in order to simplify calculations. To get it, first we assume the average temperature is the arithmetic mean of the inlet and outlet temperatures:

$$T_{avg} = \frac{T_3 + T_4}{2} \tag{21}$$

Then, considering the pressure drop is negligible for this sizing calculation ($P_4 \approx P_3$) and applying the ideal gas law $P = \rho R_{gas} T$, the average density is deduced to be:

$$\rho_{avg} = \frac{P_3}{R_{gas} \cdot T_{avg}} \tag{22}$$

where R_{gas} is the specific gas constant ($\approx 288\,\mathrm{J/kgK}$, thus using the usual value of 287 would not yield numerically incoherent results).

3.7 Point 7: Heat Density

Heat density will measure the combustion intensity, and its formula can be directly inferred by the requested units alone:

- Volumetric Intensity (I_v): Energy release per unit volume.

$$I_v = \frac{\dot{m}_f \cdot LHV}{V_{comb}} \tag{23}$$

- Pressure-Normalized Intensity (I_p): Which is the previously obtained value normalized by inlet pressure.

$$I_p = \frac{I_v}{P_3} \tag{24}$$

4 Results and Observations

4.1 Fuel Lower Heating Value

Employing the derived reaction balance ($x = 16.75$) and the standard enthalpies, the heat of reaction is found to be $\Delta H_c = -6766\,\mathrm{kJ/mol}$. Obtaining the molar mass of the fuel (M_{fuel}= **155.184g/mol**), we can derive the Lower heating value: **LHV= 43.60MJ/kg**

4.2 Fuel Flow Rates

The resulting fuel flow rates are shown in the table below:

Table 1: Fuel Flow Rates vs. Thrust

Thrust	$\dot{m}_{air}$ **[kg/s]**	T_3 **[K]**	T_4 **[K]**	$\dot{m}_f$ **[kg/s]**
Idle	20.0	500	950	0.2788
50 kN	30.0	575	1050	0.4476
100 kN	45.0	650	1225	0.8093
200 kN	77.0	750	1450	1.6848
300 kN	99.0	825	1650	2.5461

4.3 Equivalence Ratios

The stoichiometric FAR was calculated as 0.0672, taking the previously calculated fuel rates with the available air mass rate:

Table 2: Overall Equivalence Ratios

Thrust	$\dot{m}_{air}$ **[kg/s]**	$\dot{m}_f$ **[kg/s]**	**Actual FAR**	$\phi_{overall}$
Idle	20.0	0.2788	0.0139	0.2075
50 kN	30.0	0.4476	0.0149	0.2221
100 kN	45.0	0.8093	0.0180	0.2677
200 kN	77.0	1.6848	0.0219	0.3258
300 kN	99.0	2.5461	0.0257	0.3829

4.4 Air Distribution (300 kN Case)

By taking into account the partial combustion occurring in the rich zone ($FAR_{stoich} \approx 0.1001$) and obtaining the quench air based on the unburnt fuel carried over (yielding $\approx$ 0.85 kg/s), the flow splits determined are displayed below (giving them also in percentage form, for better visualization):

Table 3: Calculated Air Mass Flow Distribution (Thrust = 300 kN)

Zone	**Mass Added [kg/s]**	**% of Total**	**Equivalence Ratio**
Rich Burn Zone (Total)	**16.96**	**17.1%**	$\phi = 1.5$ (Partial)
Snout Air	*11.88*	*12.0%*	
Rich Holes (X_1)	*5.08*	*5.1%*	
Quench Zone (X_2)	**18.04**	**18.2%**	$\phi = 0.7$ (Complete)
Dilution Zone (X_3)	**63.99**	**64.6%**	
Total Air Flow	**99.00**	**100%**	

It is clear in the analysis that the Rich Zone consumes 17.1% of the total air, and the Quench zone requires approximately 18.2% of the air to oxidize the remaining fuel to lean conditions. This leaves a notable percentage of the air ($\approx$ 64.6%) for the Dilution zone only, ensuring an effective temperature acclimatization before the turbine.

4.5 Adiabatic Flame Temperatures

The resulting flame temperatures are listed below:

Table 4: Adiabatic Flame Temperatures

Zone	**Temp** (T_{ad}) **[K]**
Rich Zone	2281.6
Quench Zone	2276.6
Exit	1688.2

4.6 Combustor Volume

Using the 300 kN data ($\dot{m}_{total} = 101.55$ kg/s, computed taking into account the added fuel, $\rho_{avg} = 11.91$ kg/m^3) and $\tau = 6$ ms:

$$V_{comb} = 8.52\,\frac{m^3}{s} \times 0.006\,s \approx 0.0511\,\mathrm{m}^3 = 51.1\,L \tag{25}$$

4.7 Heat Density

Their values are compiled on the table below:

Table 5: Heat Intensity Parameters at Different Thrust

Thrust Setting	m_f **(kg/s)**	Q **(MW)**	I_v **(MW/m^3)**	I_p **(MW/m^3/bar)**
Idle	0.2788	12.16	237.88	47.58
50 kN	0.4476	19.52	381.91	38.19
100 kN	0.8093	35.29	690.52	39.46
200 kN	1.6848	73.46	1437.52	47.92
300 kN	2.5461	111.01	2172.41	51.12

5 Conclusion

The analysis, among others, sized the combustor volume at 51.1 Liters to ensure stable combustion at a full thrust of 300 kN with the requested residence time of 6 ms. The RQL mode of operation properly accounts for partial oxidation in the primary zone, directing 17.1% of the total air to the Rich Zone ($\phi = 1.5$) to minimize NO_x formation and thus reduce emissions. Subsequently, 18.2% of the airflow goes to the Quench Zone to oxidize the remaining unburnt fuel, ensuring a complete combustion.

Notably, most of the airflow (64.6%) is reserved for the Dilution Zone. This massive dilution is paramount in reduction of the gas temperature from a peak of 2281.6 K in the rich core down to 1688.2 K at the turbine inlet (T_4), as designed for the protection of downstream components. This value is slightly different from the corresponding one in the provided plots ($\approx 30K$) due to accumulated error. Furthermore, the combustor counts with a high volumetric heat intensity of 2172.41 MW/m^3 for take-off conditions, confirming the viability of the annular design using the requested Sustainable Aviation Fuel ($C_{11}H_{23}$).